2-3-26 LAB 2

① FCFS:

```
#include <stdio.h>
struct process{
    int pid;
    int at;
    int bt;
    int ct;
    int wt;
    int tat;
    ~~int completed;~~
};

void fcfs(struct process p[], int n){
    int time = 0;
    float total_tat = 0, total_wt = 0;
    printf("FCFS Scheduling\n");

    for(int i=0; i<n-1; i++){
        for(int j = i+1; j<n; j++){
            if(p[i].at > p[j].at){
                struct process temp = p[i];
                p[i] = p[j];
                p[j] = temp;
            }
        }
    }

    for(int i=0; i<n; i++){
        if(time < p[i].at){
            time = p[i].at;
        }
        time += p[i].bt;
        p[i].ct = time;
        p[i].tat = p[i].ct - p[i].at;
        p[i].wt = p[i].tat - p[i].bt;
```

```
        total_wt += p[i].wt;
        total_tat += p[i].tat;
    }
    printf("PID\t AT\t BT\t CT\t WT\t TAT\n");
    for(int i=0; i<n; i++){
        printf("%d \t %d \t %d \t %d \t %d \t %d \n",
                p[i].pid, p[i].at, p[i].bt, p[i].ct,
                p[i].wt, p[i].tat);
    }
    printf("Avg. waiting time: %.2f \n", total_wt/n);
    printf("Avg. tat : %.2f\n", total_tat/n);
}
int main(){
    int n;
    printf("Enter no. of processes: ");
    scanf("%d", &n);
    struct process p[n];
    for(int i=0; i<n; i++){
        printf("\n Process %d \n", i+1);
        p[i].pid = i+1;
        printf("Arrival time:");
        scanf("%d", &p[i].at);
        printf("Burst Time:");
        scanf("%d", &p[i].bt);
    }
    fcfs(p,n);
    return 0;
}
```

O/P: Enter no. of processes : 3

Process 1.

Arrival time: 0

Burst time: 2

Process 2:

Arrival time: 3

Burst time: 1

Process 3:

Arrival time: 5

Burst time: 5

FCFS Scheduling:

PID	AT	BT	CT	WT	TAT
1	0	2	2	0	2
2	3	1	4	0	1
3	5	5	10	0	5

Avg. waiting time = 0.00

Avg. turn around time = 2.67